

A character becomes a joint unit eigenvector

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_unit_eigenvector_of_character_ne_infinity`

theorem

Transfers a normalized GNS eigenvector to the given representation through a unitary with a specified direction.

Statement

Let A be a nonzero complex C^* -algebra, with no unit assumed and. Let H be a complex Hilbert space and $\pi : A \rightarrow \mathcal{B}(H)$ a $*$ -representation. Assume π is nonzero irreducible and that every nonzero irreducible representation of A is unitarily equivalent to it. Let $\widehat{\pi}(\lambda, a) = \lambda I_H + \pi(a)$ be its unital extension to $\widetilde{A} = \mathbb{C} \oplus A$. Write $\iota(a) = (0, a)$ and $\chi_\infty(\lambda, a) = \lambda$. Let $D \subseteq \widetilde{A}$ be a norm-closed unital $*$ -subalgebra and χ a character of D distinct from $\chi_\infty|_D$. Then there is $\eta \in H$ such that

$$\|\eta\| = 1, \quad \widehat{\pi}(d)\eta = \chi(d)\eta \quad (d \in D).$$

Assumptions

No separability of H is assumed, and D is not assumed commutative. The existence of the specified character of D is part of the input.

Conclusion

One unit vector is a joint eigenvector for every $d \in D$, with eigenvalues given by the very same prescribed character χ .

Proof route

Extend the character to a pure state; prove that its GNS restriction is nonzero; use the singleton comparison and the zero GNS norm of a centered element.

▼ Proof steps

1. Apply Pure-state extension of the prescribed character to this closed unital $*$ -subalgebra D of the nonzero unital $\widetilde{A}$ and its character χ . Obtain a pure state φ on $\widetilde{A}$ with $\varphi|_D = \chi$. Let (H_φ, ρ_U, ξ) be its GNS representation, where $\|\xi\| = 1$, and restrict it by $\rho(a) = \rho_U(\iota(a))$.
2. If $\rho(a) = 0$ for every $a \in A$, the GNS identity would give $\varphi(\iota(a)) = \langle \xi, \rho(a)\xi \rangle = 0$. Complex linearity and normalization would then give

$$\varphi(\lambda, a) = \lambda = \chi_\infty(\lambda, a).$$

Restricting to D contradicts $\chi \neq \chi_\infty|_D$. Thus ρ is nonzero. Purity gives irreducibility of ρ_U by Pure-state GNS irreducibility. Reducing subspaces for ρ also reduce $\lambda I + \rho(a)$, so ρ is irreducible.

3. Apply the singleton condition to this nonzero irreducible ρ to obtain a unitary equivalence. Extending it to the unitizations supplies a unitary $V : H \rightarrow H_\varphi$ with

$$V\widehat{\pi}(z) = \rho_U(z)V \quad (z \in \widetilde{A}).$$

We use this chosen V for the rest of the argument. It is unnecessary to identify it with a separately chosen earlier intertwiner.

4. For $d \in D$, put $q = d - \chi(d)1 \in D$. Multiplicativity and preservation of adjoints give $\chi(q) = 0$ and $\chi(q^*q) = 0$. Since the extension agrees with χ on D , the GNS identity yields

$$\|\rho_U(q)\xi\|^2 = \varphi(q^*q) = \chi(q^*q) = 0.$$

Therefore $\rho_U(d)\xi = \chi(d)\xi$.

5. Set $\eta = V^{-1}\xi$. Unitarity gives $\|\eta\| = \|\xi\| = 1$. Substituting this same vector into the intertwining equation gives

$$V\widehat{\pi}(d)\eta = \rho_U(d)\xi = \chi(d)\xi = V(\chi(d)\eta).$$

Cancel the injective V to obtain the stated joint eigenvector equations.

Main citations

- Exact theorem and proof
- Pure-state extension of the prescribed character
- Pure-state GNS irreducibility
- Singleton comparison
- The scalar character

Lean source signature (exact)

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theorem exists_unit_eigenvector_of_character_ne_infinity [Nontrivial A]
  (pi : NonUnitalCStarRepresentation A H)
  (hsingle : IsSingletonIrreducibleModel.{u, v, u} pi)
  (D : StarSubalgebra C (Unitization C A))
  [IsClosed (D : Set (Unitization C A))]
  (chi : WeakDual.characterSpace C D)
  (hchi : chi ≠ infinityCharacterOn (A := A) D) :
  ∃ eta : H, ‖eta‖ = 1 ∧
    ∀ d : D, pi.unitization (d : Unitization C A) eta = chi d • eta
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In the source	Mathematical meaning
<code>[Nontrivial A]</code>	The algebra is nonzero: $A \neq \{0\}$. This condition does not say whether a unit is assumed; that information comes from the surrounding C^* -algebra structure.
<code>(pi : NonUnitalCStarRepresentation A H)</code>	The specified $*$ -representation $\pi : A \rightarrow \mathcal{B}(H)$; no unit-preservation equation is required.
<code>(hsingle : IsSingletonIrreducibleModel.{u, v, u} pi)</code>	The specified representation is nonzero irreducible, and every nonzero irreducible $*$ -representation of the same algebra is unitarily equivalent to it.
<code>(D : StarSubalgebra C (Unitization C A))</code>	A unital complex $*$ -subalgebra D of $\tilde{A}$; commutativity is not a binder.
<code>[IsClosed (D : Set (Unitization C A))]</code>	D is norm closed in the unitization.
<code>(chi : WeakDual.characterSpace C D)</code>	The prescribed nonzero complex character χ on D .
<code>(hchi : chi ≠ infinityCharacterOn (A := A) D)</code>	χ differs from $(\lambda, a) \mapsto \lambda$ restricted to D .
<code>∃ eta : H, eta = 1 ∧</code>	One vector η is chosen in the original H , and its norm is one.
<code>∀ d : D, pi.unitization (d : Unitization C A) eta = chi d • eta</code>	For every element of D , view it in $\tilde{A}$ and apply $\hat{\pi}(d)$ to this same η ; the result is the scalar $\chi(d)$ times η .

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration:

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